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Roll No

EX-701 (GS)
B.E. VII Semester Examination, June 2020
Grading System (GS)
Power System - II
Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. Draw complete block diagram for the load frequency control of an isolated power system and explain in brief each block.

OR

Two generator rated at 120 MW and 250 MW are operating in parallel. The governor setting on the machines are such that they have 4% and 3% droop. Determine the load taken by each machine for a total load of 200 MW.

2. What are the different classical methods for voltage control in power system? Explain in brief any two of them.
3. Explain the load frequency problems in power system and explain types of frequency regulation used in power system.
4. Explain clearly with a flow chart the computational procedure for load flow solution using Gauss seidel method when system contains all types of buses.
5. What is available transfer capability of transmission network? Discuss the factors influencing available transfer capability with justification.
6. What is equal area criterion? Discuss its application and limitation in the study of power system stability.
7. Two turbo-alternators rated for 110 MW and 210 MW have governor drop characteristics of 5% from no load to full load. They are connected in parallel to share a load of 250 MW. Determine the load shared by each machine assuming free governor action.

OR

What are the factors affecting the steady-state and transient stability of the power system networks? Derive swing equation and mention its applications in power system.

8. Write short notes on (any two):
- a) Economic dispatch
 - b) Comparison of different power flow studies
 - c) Load damping
 - d) Power system restructuring
